Welcome to Code Crunch!

C5|Crunch Labs: Crunch AI Data Science

MON & WED 1-2PM Weekly | Spring 2025

Attendance

Earn a Certificate of Completion by attending at least 70% of the workshops

linktr.ee/CODE.CRUNCH

Unit #5

Regression

C5|Crunch Labs: Crunch AI Data Science

Supervised Learning

What is Supervised Learning?

- Supervised learning is training a model to predict a labeled output feature based on input features
    - e.g., dog breed, car/truck, dog/cat, …

When is an instance labeled?

- an instance is labeled if the output feature’s value is known for that instance
    - e.g., instances have a distinction for

why?

- The purpose is to predict the output feature value for unlabeled instances
    - e.g, instances that do not have the distinction for dog breed, dog/cat, car/truck

data_set - DataFrame

Index	YearsExperience	Salary
0	1	32383
1	1.1	45207
2	1.3	39751
3	2	43525
4	2.2	39891
5	2.7	56642
6	3	60150
7	3.2	54445
8	3.2	64445
9	3.7	57189
10	3.9	63218
11	4	55794
12	4	56957
13	4.1	57081

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Regression

What is Regression?

- a type of supervised learning task where the goal is to predict a numerical label or output feature
  - i.e. find a mathematical function that represents the data

Dependent Variable:

- This is the output or target you want to predict. It's continuous, such as house prices, temperature, or sales revenue.

Independent Variables:

- These are the inputs or features that influence the dependent variable. For instance, in predicting house prices, features might include the size of the house, number of rooms, and location.

data_set - DataFrame

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Types of Regression

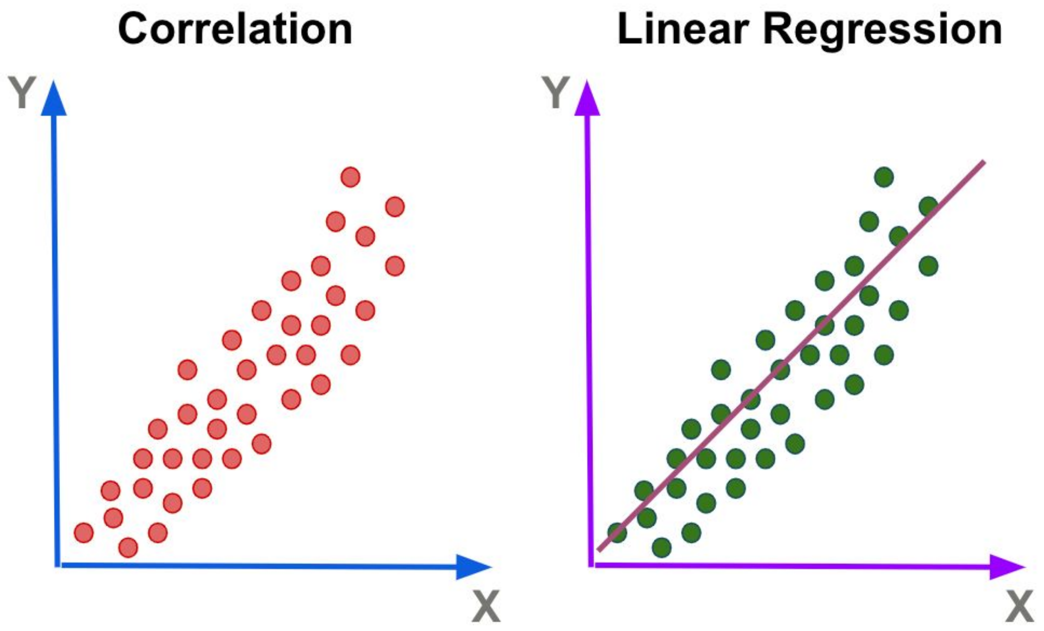
Linear Regression

- Models the relationship between variables by fitting a straight line (linear relationship).
    - Logistic Regression
    - Ridge, Lasso, and Elastic Net Regression
    -

Non-Linear Regression

- Used when the relationship between variables is not linear. These models capture more complex patterns.
    - Polynomial Regression
    - Neural Networks

$$f(x) = w_1 x_1 + \ldots + w_n x_n$$



loss function

What is a loss function

- mathematical functions that measure the difference between the predicted output of a machine learning model and the actual target value.

Common loss functions

- Mean Absolute Error
- Mean Squared Error - larger errors are given more weight
- Ridge Loss - alot of important tiny features, non-zero parameters
- Lasso Loss - some unimportant features, zeroed parameters
- Elastic Net Loss

Q&A Session

CODE CRUNCH

Attendance

Join us for the upcoming weekly sessions this Spring 2025!
(Jan 6 - Apr 4, 2025)

Monday: C5 1PM / **C3** 2PM
Tuesdays: C7 1PM
Wednesdays: C5 1PM / **C1** 2PM
Thursdays: C6 1PM / **C4** 2PM / **C8** 3PM
Friday: C2 1PM / **C9** 2PM / **C10** 3PM

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Your feedback helps us improve. Share your thoughts!
Go to our website: go.fiu.edu/codecrunch